Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/18/2016
PI Name:	Delp, Michael Ph.D.		
Project Title:	Disuse Osteopenia: A Potential Vascular Coupling Mechanism		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	09/01/2014	**End Date:**	08/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per F. Hernandez/ARC (Ed., 4/6/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Matthew Ph.D. (Indiana University School of Medicine) Delp, Judy Ph.D. (Florida State University)		
Grant/Contract No.:	NNX14AQ57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

NOTE: Continuation of "Disuse Osteopenia: A Potential Vascular Coupling Mechanism," grant #NNX12AL41G. Disuse osteopenia, which is associated with habitation in a microgravity environment, is a condition of low bone density resulting in increased fragility and greater likelihood for bone fracture. Although this condition is largely attributed to the direct effects of unloading of the bone, coupling mechanisms associated with diminished bone and marrow perfusion, attenuated fluid flow within the bone, and fewer signaling molecules from vascular cells may also play a modulating role on the bone loss associated with spaceflight.

In the tail-suspended hindlimb-unloaded (HU) rat, an animal model of microgravity-induced osteopenia, hindlimb bone and marrow blood flow decrease with acute unloading (10 min) and continue to progressively decline through 28 days of unloading. The initial and chronic decrements in perfusion are primarily due to increases in osseous vascular resistance. In addition, when the HU rat hindlimb skeleton is reloaded during standing, long bone and marrow perfusion are lower and vascular resistance higher than that in standing control animals. These higher vascular resistances in HU rats could result from 1) increased vasoconstrictor responses of bone resistance arteries, or 2) decreased endothelium-dependent and/or endothelium-independent vasodilator responsiveness of bone resistance arteries.

Specific Aims:

Aim 1. To determine whether hindlimb unloading alters the responsiveness of the principal nutrient artery (PNA) to myogenic and vasoconstrictor stimuli.

Aim 2. To evaluate the contribution of endothelium-derived NO (nitric oxide) and vasodilator prostaglandins to acetylcholine (ACh), bradykinin (BK), and flow-induced vasodilation in the femoral PNA from control and HU rats.

Aim 3. To determine whether hindlimb unloading alters the expression of endothelial NOS (eNOS), COX-1 and COX-2 mRNA and protein, as well as M2-receptor (ACh) and B2-receptor (BK) mRNA, in the femoral PNA.

Aim 4. To determine whether hindlimb unloading alters the PNA responsiveness to endothelium-independent vasodilator stimuli.

Aim 5. To determine whether hindlimb unloading, radiation exposure, and the combined effects of these treatments alter endothelium-dependent vasodilation through NO or PGI2 signaling pathways, and to determine whether this is related to changes in bone volume and indices of cancellous bone micro-architecture.

The general hypothesis of this proposal is that skeletal unloading induces alterations in skeletal resistance artery function; these alterations serve to diminish bone perfusion and ultimately disrupt mechanisms for maintaining the normal balance between bone formation and bone resorption. The specific purpose is to elucidate vasoconstrictor and vasodilator mechanisms involved in the control of bone and marrow perfusion, including endothelial signaling pathways forming nitric oxide and prostacyclin, in bone during conditions of unloading, and to determine whether deficits in bone blood flow and endothelium-dependent vascular signaling are related to unloading-induced bone loss. To accomplish these goals, the principal nutrient artery of the femur from control and 14 day HU rats will be isolated and cannulated for in vitro experimentation. Mechanisms of smooth muscle and endothelial cell function in the osseous circulation will be determined, as well as how skeletal unloading alters these vasoconstrictor and vasodilator control mechanisms (Aims 1-4). Studies will then be conducted using adenoviral gene transfer (Aim 5) to manipulate bone and marrow perfusion and endothelium-dependent vasodilation to determine whether blood flow and vascular endothelial cell signaling is coupled to disuse-associated bone remodeling in the femur (i.e., bone formation rates, mass, total, cortical and cancellous bone mineral density, and mechanical strength). These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders that occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders that occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.

Task Progress:

Three manuscripts were published in the past year with support derived from this grant. In the case of the first published manuscript (Ghosh et al., Bone 83, 2016), the purpose of this investigation was to determine the effects of spaceflight on the STS-131 and STS-135 Space Shuttle missions on mandibular bone density and bone volume. The results demonstrate that mandibular bone density is not affected, whereas mandibular bone volume is reduced by spaceflight. These findings demonstrate that non-load bearing bones are altered in a weightless environment. Although it has long been hypothesized that spaceflight-induced increases in skull bone mineral density and bone volume in humans and rats are the result of cephalic fluid shifts associated with a weightless environment, results from this study of mandibles combined with the reported increase in bone mineral density of the calvaria from the same animals suggest that other factors associated with spaceflight (e.g., elevated CO2 levels, vascular adaptations, elevated intracranial pressure) may also be important in modulating the remodeling process of non-load bearing bones in the head.
The second manuscript (Prisby et al., J. Appl. Physiol. 120, 2016) published demonstrates that hindlimb unloading (HU), low-LET ionizing radiation (Rad), and the combined effects of these two treatments (HU-Rad) diminishes endothelium-dependent vasodilation, and that both endothelium-dependent vasodilation and the NO-mediated portion of endothelium-dependent vasodilation are significantly correlated with changes in bone volume and indices of cancellous bone micro-architecture.

The third manuscript (Ghosh et al., Rad. Res. 185, 2016) addresses whether simulated weightlessness and space-relevant radiation interact to alter vascular responsiveness and bone properties. The data demonstrate that hindlimb unloading (HU) and high-LET radiation (Rad) each individually diminishes endothelium-dependent vasodilation, and the combined effects of these two treatments (HU-Rad) interact to impair endothelium-dependent vasodilation to a greater extent than either treatment does individually. With HU this reduction in endothelium-dependent vasodilation appears to be mediated by a decrease in eNOS protein levels in the vasculature, whereas Rad works through a decrease in SOD levels and an increase in XO protein levels. Further, decrements in endothelium-dependent vasodilation are

	significantly correlated with changes in bone volume, and this association between the vascular endothelium and bone loss is much stronger than the association between muscle atrophy and bone loss.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Ghosh, P, Stabley JN, Behnke BJ, Allen MR, Delp MD. "Effects of spaceflight on the murine mandible: Possible factors mediating skeletal changes in non-weight bearing bones of the head." Bone. 2016 Feb;83:156-61. Epub 2015 Nov 9. http://dx.doi.org/10.1016/j.bone.2015.11.001 ; PubMed PMID: 26545335 , Feb-2016
Articles in Peer-reviewed Journals	Prisby RD, Alwood JS, Behnke BJ, Stabley JN, McCullough DJ, Ghosh P, Globus RK, Delp MD. "Effects of hindlimb unloading and ionizing radiation on skeletal muscle resistance artery vasodilation and its relation to cancellous bone in mice." J Appl Physiol (1985). 2016 Jan 15;120(2):97-106. Epub 2015 Oct 15. http://dx.doi.org/10.1152/japplphysiol.00423.2015 ; PubMed PMID: 26472865 , Jan-2016
Articles in Peer-reviewed Journals	Ghosh P, Behnke BJ, Stabley JN, Kilar CR, Park Y, Narayanan A, Alwood JS, Shirazi-Fard Y, Schreurs A-S, Globus RK, Delp MD. "Effects of high-LET radiation exposure and hindlimb unloading on skeletal muscle resistance artery vasomotor properties and cancellous bone microarchitecture in mice." Radiat Res. 2016 Mar;185(3):257-66. http://dx.doi.org/10.1667/RR4308.1 ; PubMed PMID: 26930379 , Mar-2016